MATH 241: Calculus III

Fall 2026

Lecture 23: Green's Theorem

Background & Motivation

Green's Theorem is the first of the three big theorems in vector calculus. It converts a line integral around a closed curve into a double integral over the enclosed region. This is powerful both computationally—sometimes one side is much easier than the other—and theoretically, as it connects circulation to curl.

1. Green's Theorem

– Converts a line integral around a closed curve into a double integral over the enclosed region (and vice versa)

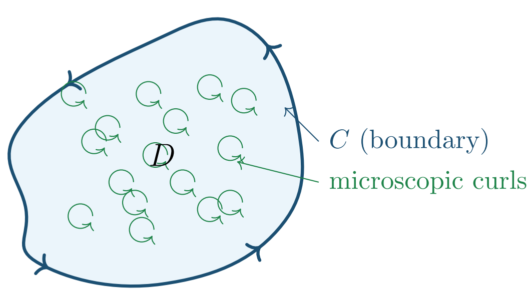
Theorem 1: Green's Theorem (Circulation Form)

Let C be a positively oriented (CCW), simple, closed curve bounding a region D. If P and Q have continuous partial derivatives on an open region containing D:

$$\boxed{\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA}$$

Requirements:

- C must be ________________________
- P, Q must be _____________________________



Macroscopic circulation around C = sum of all microscopic curls $(Q_x - P_y)$ inside D.

Theorem 2: Green's Theorem (Vector Form)

Recall that $\oint_C P\,dx + Q\,dy$ is really a line integral of $\mathbf{F} = \langle P, Q\rangle$ along C. Writing Green's Theorem entirely in terms of $\mathbf{F}$:

$$\boxed{\oint_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}\,'(t)\,dt \;=\; \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA}$$

The middle form is what we compute directly (parameterize C, dot, integrate). The right side is what Green's lets us swap to—a double integral over D.

Preview: The quantity $Q_x - P_y$ is what we will call the **curl** of $\mathbf{F}$ (next lecture). It measures local rotation. When we later learn **Stokes' Theorem**, we will see that Green's is just Stokes' with the surface flat in the xy-plane. Swap $Q_x - P_y$ for curl and you get Stokes'.

When to use Green's Theorem

- Hard line integral but easy double integral $\Longrightarrow$ use Green's left-to-right
- Hard double integral but easy line integral $\Longrightarrow$ use Green's right-to-left
- To compute areas via line integrals

Example 1: Verification on a Triangle (Both Sides)

Verify Green's Theorem for $\oint_C (x^2y\,dx + xy^2\,dy)$ around the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.

Example 2: Using Green's to Simplify a Circle Integral

Evaluate $\displaystyle\oint_C (e^x + y^3)\,dx + (x^3 - \sin y)\,dy$ where C is the unit circle (CCW).

2. Area via Green's Theorem

Area Formulas via Green's Theorem

Setting $Q_x - P_y = 1$ in Green's Theorem gives area formulas. After parameterizing C with $\mathbf{r}(t) = \langle x(t), y(t) \rangle$:

$$\text{Area}(D) = \frac{1}{2} \int_a^b \left(x \frac{dy}{dt} - y \frac{dx}{dt} \right) dt$$

Also written in differential notation: $\frac{1}{2} \oint_C (x\,dy - y\,dx)$, or equivalently $\oint_C x\,dy$ or $-\oint_C y\,dx$.

Example 3: Area of an Ellipse

Find the area enclosed by $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ using Green's Theorem.

3. When Green's Theorem Fails

– Green's requires P, Q to have continuous partials on **all** of D

– If $\mathbf{F}$ is undefined inside D (e.g., at the origin), the theorem does **not** apply directly

Example 4: The $1/(x^2+y^2)$ Counterexample

Compute $\displaystyle\oint_C \frac{-y}{x^2+y^2}\,dx + \frac{x}{x^2+y^2}\,dy$ around the unit circle. Show Green's Theorem does not apply.

Practice Problems

Problem 1. Use Green's Theorem to evaluate $\oint_C (2y\,dx + 3x\,dy)$ around the unit square $[0,1]^2$ (CCW).

Problem 2. Evaluate $\oint_C (y^2\,dx + x^2\,dy)$ where C is the boundary of the region between $y = x$ and $y = x^2$, $0 \le x \le 1$ (CCW).

Problem 3. Use Green's Theorem to find the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \le t \le 2\pi$.

Problem 4. Evaluate $\oint_C (x^2 y\,dx - xy^2\,dy)$ where C is the circle $x^2 + y^2 = 4$ (CCW).

Problem 5. True or false (explain briefly):

(a) If $Q_x - P_y = 0$ everywhere in D, then $\oint_C P\,dx + Q\,dy = 0$.

(b) Green's Theorem can be applied to $\mathbf{F} = \langle x/r^2, y/r^2 \rangle$ (where $r = \sqrt{x^2+y^2}$) on any region not containing the origin.

Answer: